



Modeling the effect of gravity on periodic alternating nystagmus

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ABSTRACT

Periodic alternating nystagmus (PAN) is a rare oscillatory ocular motor disorder. The effects of gravity on the dynamic behavior of PAN can be studied by monitoring the nystagmus while changing head orientation. Previous studies of patients with PAN reached different conclusions about the effect of changing the orientation of the head relative to gravity on the ongoing PAN, either no effect or a damping of the nystagmus within several minutes. What neuronal circuits could account for the difference in the effects of gravity among PAN patients? We modeled how the brain resolves the tilt-translation ambiguity in normal individuals and added an unstable, oscillatory vestibular system generating PAN. PAN was suppressed in our patient in ear-down positions, in a similar pattern to that of a previously reported patient. This effect was simulated by reducing the gain of the projection of the “rotation feedback” loop to the velocity-storage integrator to approximately 5% of its normal value. With normal “rotation feedback” PAN is expected to dissipate quickly as soon as the head is rotated away from upright position. Moreover, by disconnecting the rotation feedback completely (gain = zero) the model simulated PAN that was reported to be unaffected by gravity. Thus, understanding the effect of this single parameter, the gain of the rotation feedback, can explain the observed variability among our own and previous studies.

1. Introduction

Periodic alternating nystagmus (PAN) is an ocular motor disorder characterized by spontaneous horizontal nystagmus that reverses direction approximately every 2 min [1]. Previous studies of patients with PAN reached different conclusions about the effect of changing the orientation of the head relative to gravity on the ongoing PAN. In four patients tested in supine and ear-down positions PAN was unaffected by gravity [2]. In a single patient, Chung et al. found damping of the nystagmus within several minutes (approximately two periods of the PAN oscillation) in ear-down positions [3]. Here we examined the effects of gravity on an additional patient with PAN to help resolve the different results from prior works. Our strategy was to combine a model of the interaction of gravity with other vestibular signals in normal individuals [4] with a model of the mechanism underlying PAN [1]. We

then used this combined model to simulate the effect of lesions in central vestibular circuits that might account for the different effects of changing the orientation of the head relative to gravity on the ongoing PAN. The example of PAN and its interaction with gravity highlights the value of these models in interpreting and localizing complex clinical findings in patients with central vestibular disorders.

The vestibular system – the vital sixth sense that ensures stable posture and clear vision when we move – apprises the brain of the motion and position of the head with respect to the external world [5]. Angular acceleration is detected by the semicircular canals (SCCs) and linear acceleration by the otolith organs. However, as Einstein’s principle of equivalence states, any linear accelerometer (as the otolith organs) cannot distinguish changes in linear acceleration due to tilt of the head with respect to gravity from translation of the head in the environment [6]. Instead, the otoliths sense the gravito-inertial acceleration

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(GIA) vector resulting from the sum of linear acceleration due to translation and that due to gravity. The brain must solve this ambiguity because, for example, in the case of vestibulo-ocular reflexes, the response to a lateral tilt of the head is a torsional eye movement while that to a lateral translation of the head is a horizontal eye movement. While no perfect mathematical solution exists to resolve this tilt-translation ambiguity in the otolith signal, under most natural conditions the brain can use context or existing beliefs to disambiguate the GIA into its two constituents. An example using *context* is to monitor activity in the semicircular canals to detect dynamic tilt before a change in GIA is attained. An existing belief, for example, could make the brain expect that any *sustained* change in GIA comes from a tilt, and not from a translation.

Unfortunately signals from the semicircular canals are not perfect representations of head rotation as they also include inherent biological noise, their response fades to head movements of long duration, and there are unwanted motion aftereffects when the head stops moving. Thus, the brain must use a sensor fusion approach to optimally combine these two ambiguous and imperfect signals to estimate tilt, translation, and rotation. Previous studies have developed powerful models implementing these ideas quantitatively and were able to simulate many common experimental findings ([7,8–10]; [11]; [12,4,13]). Here we will extend these models to interpret the response of our patient and those previously reported from patients with PAN in whom the effects of changing the orientation of the head relative to gravity on the ongoing nystagmus were examined.

Fig. 1 shows a diagram of the main elements of the model we implemented. In this model there are four critical operators: tilt estimator, somatogravic feedback, rotation feedback and velocity storage.

The tilt estimator integrates canal inputs to provide an additional estimate of the orientation of the head with respect to gravity. Any head

rotation can be characterized by an imaginary straight line (the axis of rotation) through the head around which all other points of the head revolve. For example, the axis of pure horizontal head rotation is in the cephalo-caudal orientation. By considering the orientation of the axis of head rotation to that of the internal estimate of the pull of gravity, head rotations can be divided into two categories: parallel and non-parallel to gravity. Integration of head rotation velocity signals about an axis parallel to gravity determines the heading angle for navigation (like a ship's dead reckoning). Integration of head rotation velocity signals about an axis non-parallel to gravity, determines the tilt relative to gravity. This can be mathematically formulated using the integral of the cross-product between the previous estimated gravity vector and the head rotation velocity [11].

In long time periods, however, the integration operation becomes unreliable since the canal signals contain noise which accumulates over time. For this reason, the brain combines a tilt estimation from the otolith, commonly referred as the somatogravic feedback (Fig. 1), to the tilt estimation from the SCCs. The weighted combination of gravity estimates by SCCs (through integration of non-parallel head rotations with respect to gravity) and the otolith (through low-pass filtering of the GIA) usually generates a reliable estimate of gravity over both short and long time periods. In the case of sustained translation, however, the nature of the somatogravic feedback reflects the belief that any *sustained* change in GIA must come from a tilt, and not from an extended translation. This results in an erroneous interpretation of tilt, also known as the somatogravic illusion [15–17,4,18].

The estimation of gravity by the central nervous system can be complicated by any “virtual” head rotation, which are velocity signals that do not correspond to an actual rotation of the head in the environment. From a computational perspective, the brain does not distinguish virtual from real head rotations. The most common virtual head

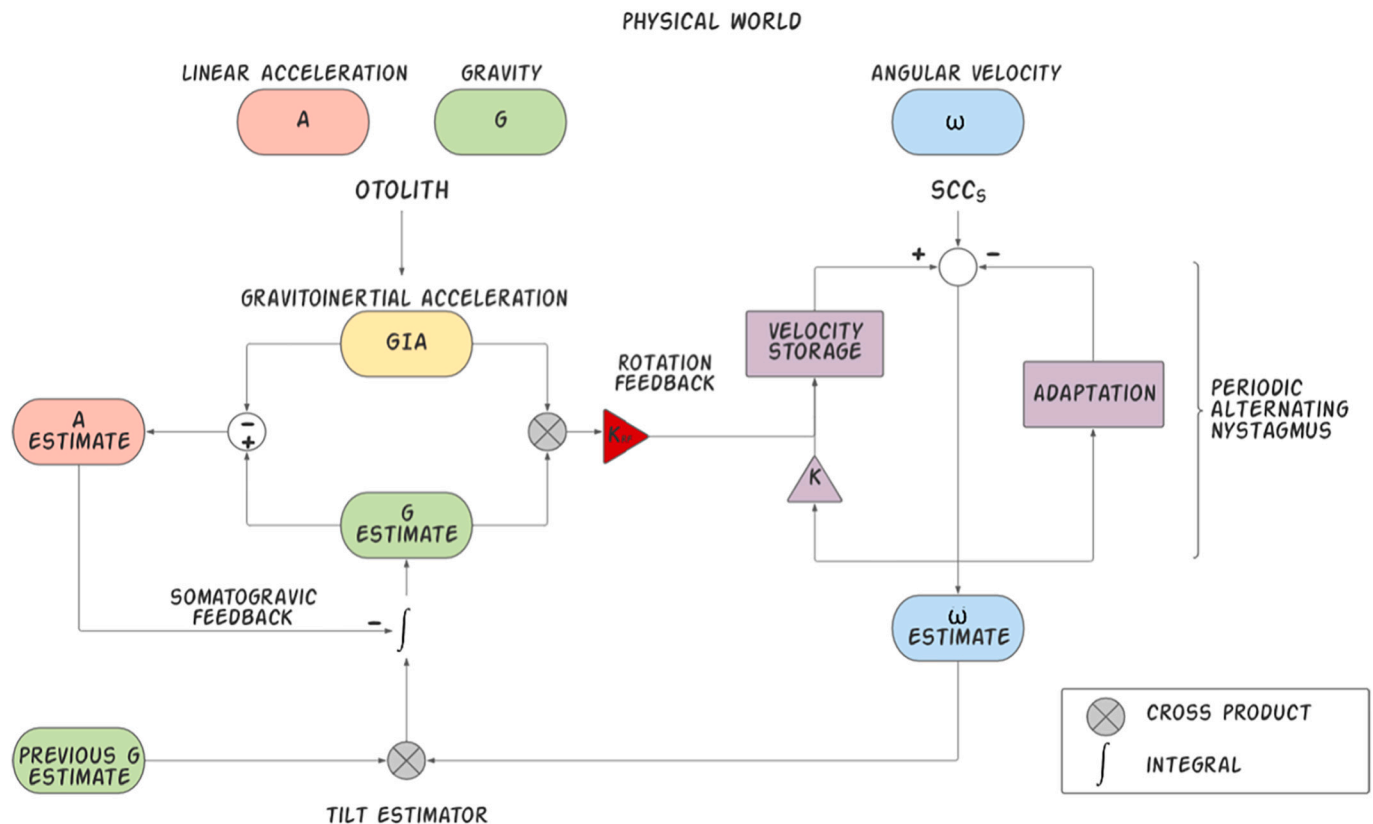


Fig. 1. Tilt translation ambiguity 3D Model. We have modified the model implemented by Glasauer and colleagues to simulate central positional nystagmus [14]. An adaptation operator has been added to the angular velocity estimation (ω). This adaptation operator, in combination with the velocity storage component can produce periodic alternating nystagmus if their parameters reach a particular combination of values.

rotation naturally arises from the activity of the SCCs during the aftereffect following a head movement. For example, after going supine from a sitting position (pitch-up), the aftereffect from the post-rotatory response SCCs signals a virtual head rotation in the opposite direction (pitch-down). The integration of virtual head rotation velocity signals about an axis non-parallel to gravity leads to an erroneous estimate of gravity. Moreover, the resulting disparity between the orientation of the GIA and the erroneous estimated gravity will lead to a false sense of translation, along the rostral-caudal axis of the body, towards the feet. Therefore, virtual head rotations, which do not correspond to actual rotations of the head, can have a destabilizing effect and lead to inaccurate vestibulo-ocular and vestibulo-spinal responses.

To minimize the effect of these virtual head rotations the brain can use the context from the otolith signals and estimated gravity direction to verify that rotation inputs are consistent with the estimated change of gravity orientation. This context comes in the form of a rotation feedback (Fig. 1) signal that offsets the virtual head rotation velocity (the input) to adjust estimated gravity until it again matches the GIA (desired output). This type of problem is called an “inverse problem” because it begins with the effects (realigning estimated gravity towards GIA), then calculates the causes (the virtual head rotation that will produce this result). Mathematically, a solution of this “inverse problem” can be formulated as the cross-product between GIA and the current estimated gravity and is named the rotation feedback. The output of this rotation feedback is incorporated into the VOR through the velocity storage integrator [19,20,21,22,23,8,13].

One method to probe the effect of rotation feedback is to measure the reduction in post-rotatory nystagmus caused by a head tilt, an effect known as “tilt-suppression” [24,25,26,27,28]. When the head is upright during the post-rotatory period, the axis of virtual head rotation is parallel to gravity and therefore has no destabilizing effect on the vestibular system (estimated gravity is not updated). If instead, the head is tilted during the post-rotatory nystagmus, the virtual rotation axis has an axis that is not parallel to gravity, leading to an erroneous estimation that the head has rotated relative to gravity. The resulting disparity between the orientation of the GIA and the erroneous estimated gravity would then activate the rotation feedback, and the output of which is incorporated into the velocity storage integrator to offset the post-rotatory nystagmus. The accelerated decay of post-rotatory nystagmus decay is called “tilt-suppression” or “dumping” of post-rotatory nystagmus.

How does this conceptual formulation apply to studying the effect of changing the orientation of the head relative to gravity on an ongoing PAN? PAN can also be interpreted as a virtual head rotation though periodically changing its direction. As such, similar considerations of the geometric relationship between the axis of virtual head rotation and gravity in the resolution of the tilt-translation ambiguity, apply to PAN.

We next turned to ablation studies of the nodulus and ventral uvula in monkeys, to look for the possible physiological and anatomical underpinnings of the mechanisms that implement these mathematical operations ([29], [30]; [31,32]). Ablation of these structures in monkeys not only causes PAN, but also impairs tilt suppression of post rotatory nystagmus, and eliminates the steady-state response during off-vertical axis (OVAR) rotation. These effects of gravity are thought to be generated through the vestibular velocity-storage integrator via the rotation feedback loop. Together with the results of the ablations studies, we hypothesized that pathological variations in the gain of rotation feedback account for the difference in the effects of gravity among PAN patients. A normal gain of rotation feedback would result in a quick suppression PAN with the head tilted down. A low gain of rotation feedback could lead to gradual (over a few minutes) suppression of PAN in head down positions. With no rotation feedback at all, PAN should be unaffected by head down positions. We used computational modeling of patient data to test our hypothesis.

2. Methods

2.1. Patient and eye movement recordings

Data was collected from the same patient and with the same methods as in our previous article [33]. The patient was a 44-year-old woman who presented with imbalance and one month of blurred vision. Examination of eye movements showed PAN. Positive anti-YO antibodies with underlying breast cancer established the diagnosis of paraneoplastic cerebellar degeneration. The patient was treated with Baclofen 20 mg daily for the nystagmus. Horizontal eye movements were recorded using a head-mounted video-oculography system (ICS Impulse, GN Otometrics, Taastrup, Denmark) with a sampling rate of 220 Hz. The Dokuz Eylül University Review Board approved the experimental protocol and written informed consent was obtained from the patient. All testing was in the light and in the sitting position one meter away from the central target on a wall with the head unrestrained. For the experiments shown here the patient was quickly moved into different positions while recordings continued. Rightward slow-phase velocities (from the patient's point of view) are positive and leftward negative.

2.2. Modeling

We implemented a model using MATLAB and Simulink (The MathWorks, Natick, MA). The model was a modified version of the one previously developed by Glasauer and colleagues to simulate central positional nystagmus [14]. This model is shown in a simplified diagram in Fig. 1 and simulates how signals coming from the otolith organs and the semicircular canals are combined in 3 dimensions to estimate the orientation of the head relative to gravity (or gravity relative to the head), the angular velocity of rotation of the head, and the linear acceleration of the head. We initially implemented the model to produce normal responses without any spontaneous nystagmus. First, we made the key modification of adding an adaptation operator to the rotational velocity processing component and modified the values of the velocity storage to produce PAN in a similar manner as in our previous study [1]. We also added a variable bias term to the angular velocity sensed by the canals to calibrate the amplitude of the PAN. Then, we studied the effect of changing the gain of the rotation feedback block on the effect of changes in orientation relative to gravity on PAN. All other elements of the model were kept intact, including the somatogravic feedback loop.

3. Results

The tilt-translation ambiguity 3D model predicts that if the estimated direction of gravity and axis of head rotation (real or virtual) are not parallel, the brain will update its current estimate of direction of gravity. In the case of virtual head rotation signals, however, this leads to erroneous information about orientation of gravity that is reflected in a misalignment of estimated gravity with GIA. This, in turn, activates the rotation feedback to minimize this misalignment. PAN may be considered as a *virtual* head rotation around a rostro-caudal axis with an intensity and direction that change periodically over time. As such, characterizing the dynamic behavior of PAN in ear-down and supine positions where the axis of PAN virtual rotations is 90 degrees with respect to gravity would enable one to explore the function of the rotation feedback in patients with this disorder.

Experiment 1 shows a recording of horizontal slow-phase velocity in a patient with PAN who quickly lies down from upright to right-ear down (Fig. 2), left-ear down (Fig. 3) or supine (Fig. 4) positions. A fast rotation feedback response would be essential to rapidly restoring an accurate estimate of gravity. The PAN in the ear-down position (Figs. 2,3,4, blue dots) in our patient, however, was suppressed much more slowly than if the rotation feedback was working normally. In other words, the tilt-suppression mechanism that is normally mediated by “rotation feedback”, was impaired. We assumed a much smaller

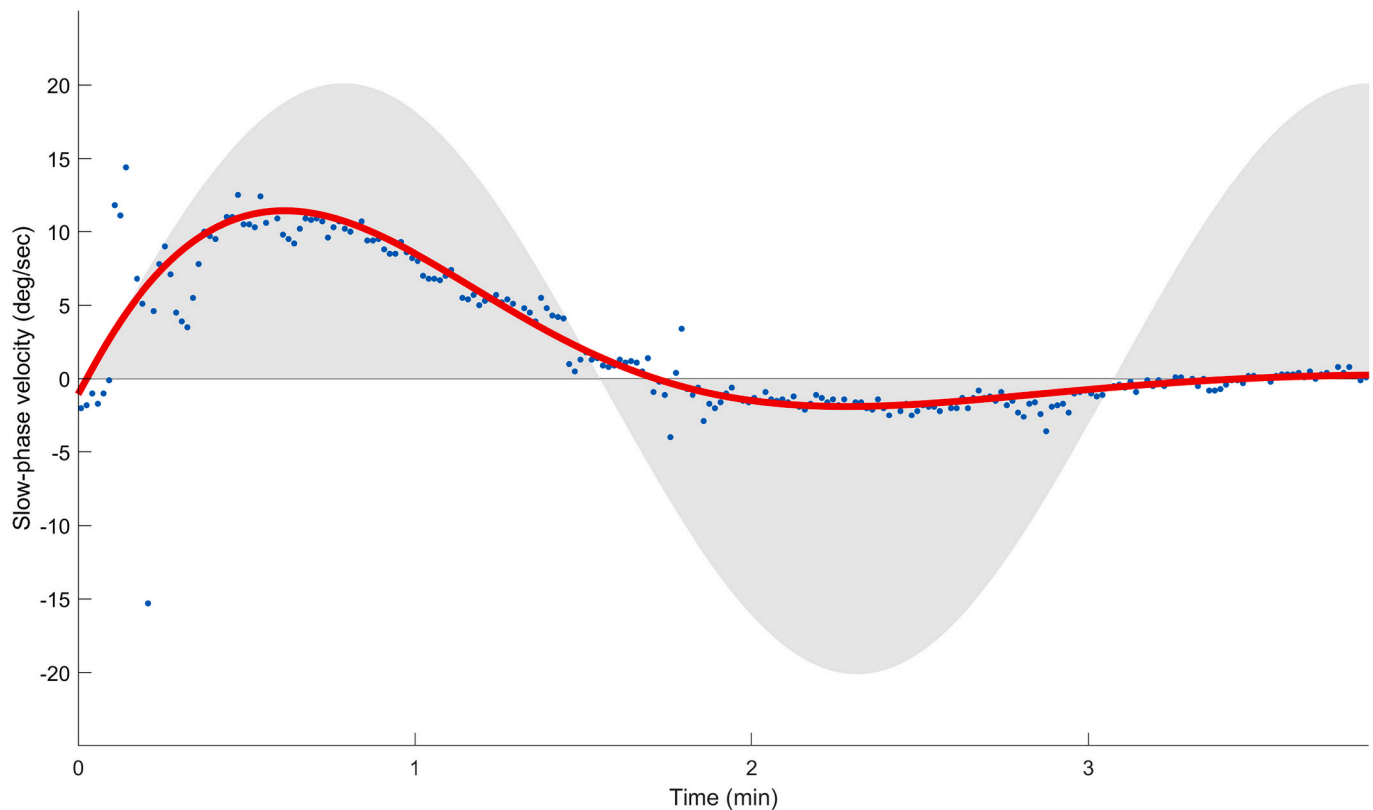


Fig. 2. Effect of left-ear down positioning on the pattern of PAN. Horizontal slow-phase velocity (deg/s) is plotted against time (minutes). At time zero, the patient was brought promptly from sitting upright, to the left ear down position by tilting the head laterally 90 degrees about the roll axis. The blue dots correspond to the patient's horizontal slow-phase velocity. The solid red line is a simulation of the yaw axis angular velocity. Slow-phase velocity of the upright simulation is shown by the envelope of the gray shaded area. Parameter values for simulation include: PAN peak amplitude 20 deg./s, rotation feedback gain 0.085 and a bias of 0 deg./s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

rotation feedback gain to account for this slower decay and accordingly modified the model parameters by using a value of the rotation feedback gain approximately 5% of its normal value. The modified model simulation of yaw-axis angular velocity closely fits the patient's data (Fig. 2, Fig. 3 and Fig. 4, red line).

To completely characterize PAN dynamics with respect to gravity, we retested our model against data from previous studies. In a single patient, Chung et al. [3] found that PAN was suppressed in ear-down positions within 7 min (approximately two periods of the PAN oscillation). As the PAN response was like that shown by our patient, we have assumed an analogous reduction in the rotation feedback gain. After extracting parameters that could describe the oscillatory behavior of the PAN shown in that paper, we adjusted the values of two model parameters, cycle duration and peak slow-phase of PAN. We simulated data from Chung et al. [3] recording of a patient with PAN, who was brought promptly from sitting upright to a left-ear down position. Simulating a model that included a rotation feedback gain value corresponding to approximately 5% of its normal value, adequately fitted their data. The conditions of this simulations are like the ones shown in Fig. 3.

In contrast, Furman et al. [2] reported that PAN was unaffected by gravity in four patients tested in supine and ear-down positions. This implies that the brain's ability to counteract the destabilizing effects of virtual head rotation signals on an estimation of gravity has been neutralized. The model predicts that the responsible neural circuit is the rotation feedback; as such we have disconnected this loop from the velocity storage circuit (gain was set to zero). Model simulation for yaw axis angular velocity is shown for left ear-down position in Fig. 5 (red line). The simulation shows that PAN was unchanged in the left-ear down position which agrees with Furman et al. report.

4. Discussion

4.1. Estimating gravity and its interaction with rotations of the head

To interpret the dynamic behavior of PAN in different positions with respect to gravity one must first consider how the brain normally estimates gravity, and how it uses that information to minimize the effects of vestibular signals that might lead to an erroneous estimate of gravity. The brain relies on a recursive approach to estimate gravity, by combining information about rotational motion from the semicircular canals and information about tilt and translational motion from the otoliths to keep updating its estimate of gravity direction. As described in the introduction, for inputs from the semicircular canals, the brain distinguishes between parallel and non-parallel axis head rotations with respect to gravity, because only the latter indicates a rotation that changes the orientation of the head relative to gravity. It then uses mathematical integration to update the estimated gravity orientation. A rotation feedback circuit, mediated through velocity storage mechanism (Fig. 1) helps to cancel any inappropriate vestibular responses.

4.2. The orientation of the head relative to gravity, rotation feedback and PAN

How do we use this formulation to analyze the effects of gravity on PAN? PAN is modeled as unstable velocity storage that in combination with a normal short-term (minutes) vestibular adaptation circuit that rebalances tonic activity in the vestibular nuclei yields oscillatory virtual head rotation signals. In ear-down and supine positions, the rotational axis of PAN becomes non-parallel with respect to gravity, which leads to erroneous estimates of gravity and linear acceleration. To counteract the

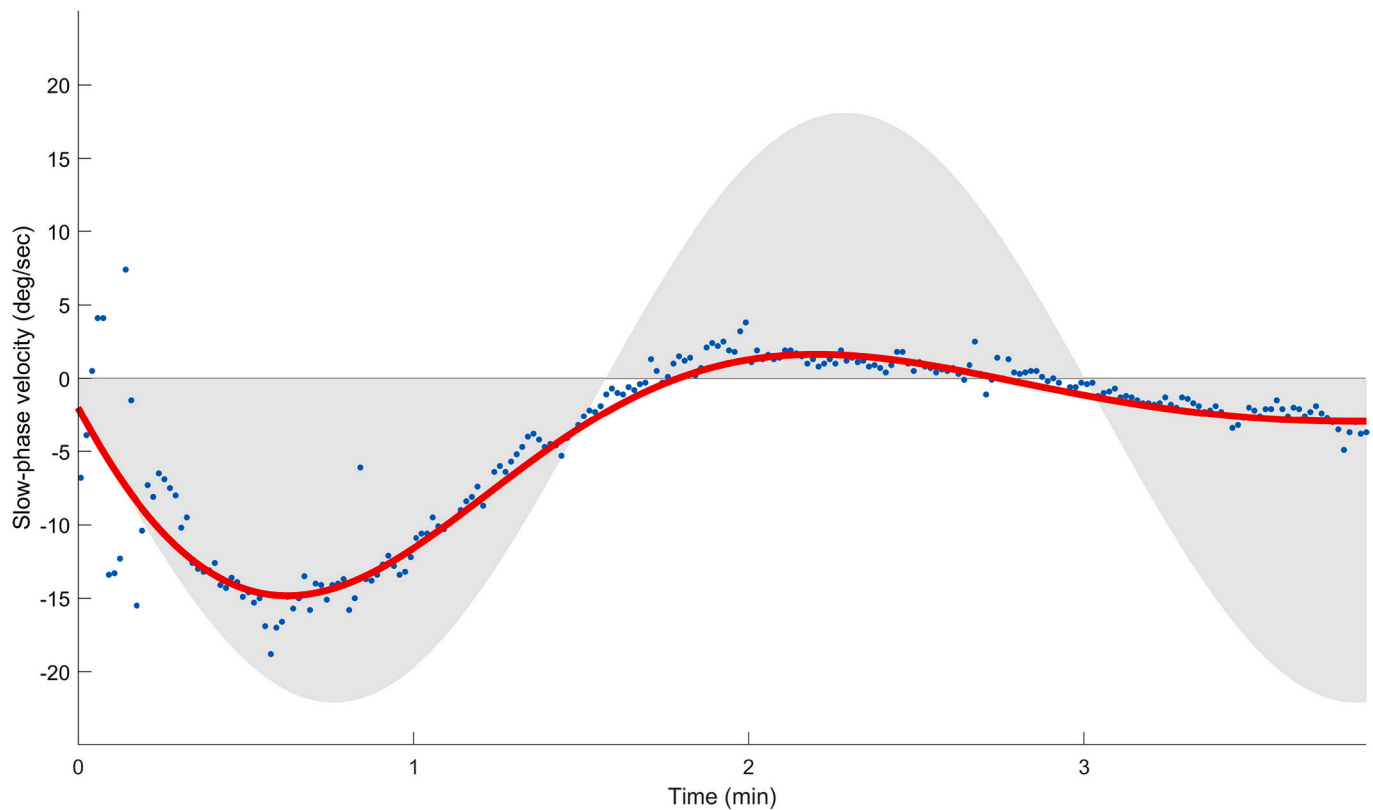


Fig. 3. Effect of right-ear down positioning on the pattern of PAN. Horizontal slow-phase velocity (deg/s) is plotted against time (minutes). At time zero, the patient was brought promptly from sitting upright, to the right ear down position by tilting the head laterally 90 degrees about the roll axis. The blue dots correspond to the patient's horizontal slow-phase velocity. The solid red line is a simulation of the yaw axis angular velocity. Slow-phase velocity of the upright simulation is shown by the envelope of the gray shaded area. Parameter values for simulation include: PAN peak amplitude 20 deg./s, rotation feedback gain 0.065, and a bias of 2 deg./s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

effect of virtual head rotation induced by PAN on estimated gravity and linear acceleration, the brain must solve the inverse problem, by computing an offsetting virtual head rotation velocity signal to adjust estimated gravity, until it matches the GIA that is signaled from the otoliths. Normally, this correction is achieved through the rotation feedback circuit. As such, examining patients with PAN in the ear-down and supine positions is analogous to “tilt-suppression of, for example, postrotatory nystagmus, which tests the underlying function of the rotation feedback circuit.

Can the performance of rotation feedback in patients with PAN be determined from previous studies? Furman et al. reported PAN was unaffected by gravity in four patients tested in supine and ear-down positions, which implied the rotation feedback was not functioning [2]. In contrast, Chung et al. described a single patient with PAN who needed a little more than two cycles (about 7 min) in ear-down positions to suppress the PAN [3]. Our patient here, showed a pattern of suppression of PAN in supine and ear-down positions like that of Cheung et al. The pattern of suppression in these two patients, implies that in some patients with PAN, the “rotation feedback” circuit can still have some effect on the velocity storage circuit, albeit slower (over minutes) and less complete (some misalignment of the inferred gravity vector and the GIA remained) than normal.

Finally, the incorporation of unstable velocity storage proposed by Leigh et al. to the tilt-translation ambiguity resolution model allowed us to make quantitative predictions of the dynamic behavior of PAN in different head positions [1]. By lowering the rotation feedback gain by 95% of its normal value, the simulations of PAN in supine and ear-down positions fitted the data from our patient and that of the previous study by Chung et al., showing the same pattern of modulation of PAN [3]. Remarkably, a very low gain of the rotation feedback was sufficient to

suppress PAN within two cycle duration. On the other hand, simulations using a zero gain of rotation feedback produced no modulation of PAN in supine and ear-down positions, in line with Furman et al. report [2].

4.3. Possible mechanisms underlying the effects of rotation feedback on PAN

Is the rotation feedback output being fed through a very low gain the only way to account for the positional observations in PAN? The rotation feedback has a strong geometric interpretation that reveals other possible computational threads to reduce the rotation feedback gain. The rotation feedback implements the vector cross product, which is the axis to rotate one vector, the estimated gravity, into the other, the GIA (Fig. 1). As the cross product is compatible with scalar multiplication (α) such that $(\alpha \cdot \vec{GIA}) \times \vec{G} = \vec{GIA} \times (\alpha \cdot \vec{G}) = (\vec{GIA} \times \vec{G}) \cdot \alpha$, employing a similar reduction in gain to any rotation feedback input (GIA or G) would be computationally equivalent to our model configuration. Moreover, just like all vectors, the cross product vector can be represented by a unit vector which defines its direction and the magnitude or length of the vector. The formula for the rotation feedback output vector $\vec{GIA} \times \vec{G} = \|\vec{GIA}\| \cdot \|\vec{G}\| \cdot \sin(\theta) \cdot \text{unit vector}$, provides an explicit expression of the cross product magnitude, $\|\vec{GIA}\| \cdot \|\vec{G}\| \cdot \sin(\theta)$ (the multiplication of the magnitudes of GIA ($\|\vec{GIA}\|$), estimated gravity ($\|\vec{G}\|$), and the sine of the angle between GIA and G). Therefore, the localization of apparent reduction to the gain of the rotation feedback can be more versatile. It can be due to (1) error in the magnitude of GIA, (2) error in the magnitude of gravity, (3) error in the estimation of the angle between GIA and gravity or (4) combination of the above. These

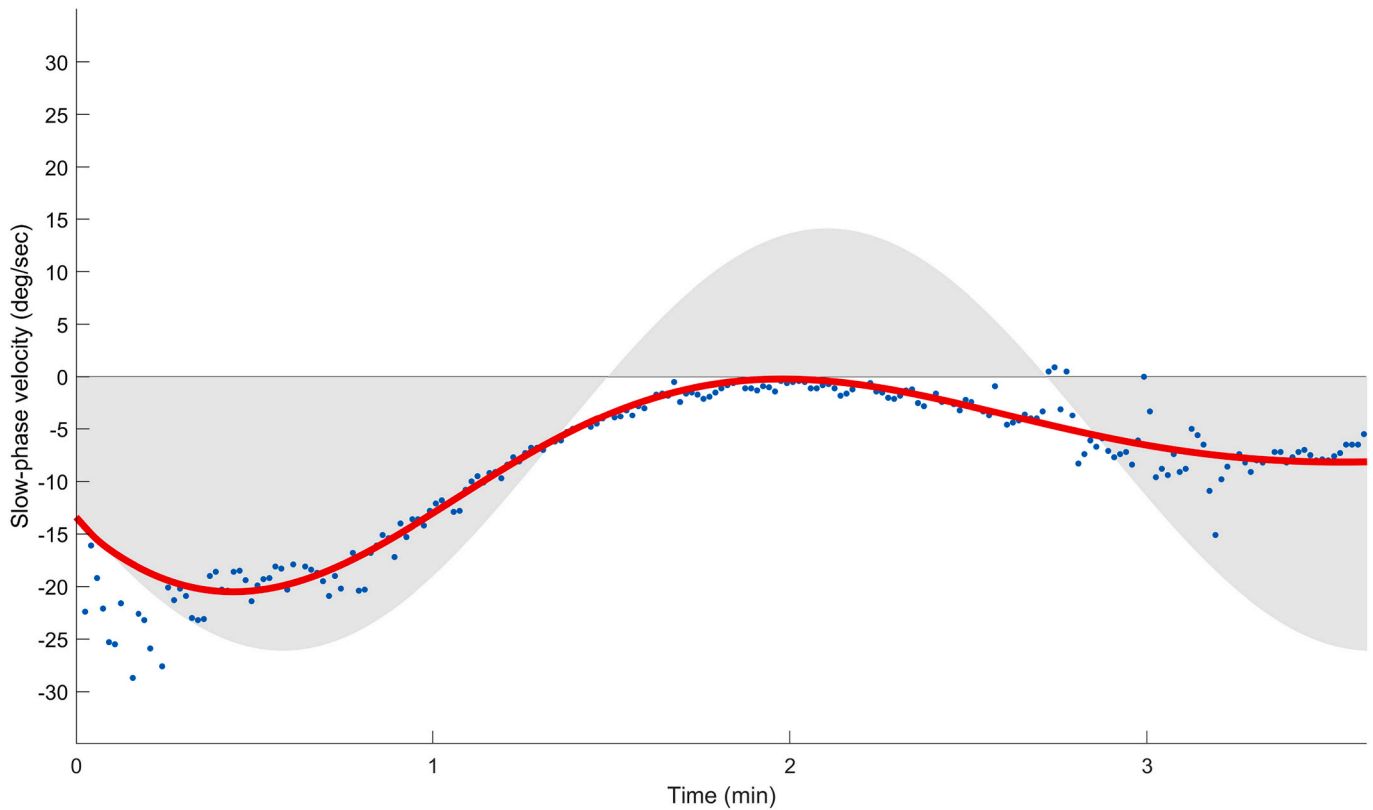


Fig. 4. Effect of supine positioning on the pattern of PAN. Horizontal slow-phase velocity (deg/s) is plotted against time (minutes). At time zero, the patient was brought promptly from sitting upright, to the supine position by tilting the head backwards 90 degrees about the pitch axis. The blue dots correspond to the patient's horizontal slow-phase velocity. The solid red line is a simulation of the yaw axis angular velocity. Slow-phase velocity of the upright simulation is shown by the envelope of the gray shaded area. Parameter values for simulation include: PAN peak amplitude 20 deg./s, rotation feedback gain 0.05, and a bias of 6 deg./s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

computational threads bear relevance to the anatomic connections of the nodulus and ventral uvula with vestibular end organs. For example, the nodulus and ventral uvula receive ample primary projections from the otolith which sense the GIA [34]. This anatomical relationship might explain the error associated with the magnitude of GIA which manifests as a lowering of the gain of the rotation feedback.

For the patients with PAN who show some residual effect of rotation feedback on the velocity storage circuit, the model can make a prediction of how to compensate for the lower gain by increasing the effect of the rotation feedback and therefore accelerate the suppression of PAN. To facilitate the interpretation of the cross product, one can express the rotation feedback cross product as a regular matrix product of a skew-symmetric matrix, with the GIA components and the G vector.

$$(GIA \times G) \cdot \alpha = \begin{bmatrix} 0 & -GIA_z & GIA_y \\ GIA_z & 0 & -GIA_x \\ -GIA_y & GIA_x & 0 \end{bmatrix} \cdot \begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix} \cdot \alpha$$

The rotation feedback for virtual yaw head rotations in the ear down positions, when GIA_z and GIA_x are both zero, is $\begin{bmatrix} 0 \\ 0 \\ -GIA_y G_x \alpha \end{bmatrix}$, and for virtual yaw head rotations in the supine position when GIA_z and GIA_y are both zero is $\begin{bmatrix} 0 \\ 0 \\ -GIA_x G_y \alpha \end{bmatrix}$. One can see that increasing the magnitude of

GIA in the axis parallel to earth gravity (GIA_y in ear down positions and GIA_x in supine position) would have a similar effect on PAN suppression to an increase in the gain of the rotation feedback. A test for this hypothesis would be to accelerate a patient with PAN in an elevator to increase GIA_y or decrease GIA_y while keeping the head in the ear down

position. The prediction would be a faster suppression of the PAN in the ear down position with increased GIA_y , that is, accelerating upwards.

4.4. A summary of the analytical approach to the dynamic interaction of PAN with gravity

In summary, the approach to analyze the dynamic behavior of PAN with respect to gravity depends on considering: (1) how the brain resolves the tilt-translation ambiguity; (2) the brain interprets a virtual head rotation as an actual head rotation in its computations; (3) interactions between the PAN and gravity can produce variation in the period and amplitude of PAN; and (4) in patients with PAN due to central lesions, primarily involving the cerebellar nodulus and ventral uvula, the rotation feedback, functions inadequately, but in different levels of performance. Accordingly, the performance of rotation feedback can be qualitatively determined at the bedside, when the rotational axis of PAN becomes non-parallel with respect to gravity, such as in supine or ear-down positions. If the PAN varies by changes in head orientation with respect to gravity, it suggests the rotation feedback can still exert its effects on the velocity storage mechanism. For example, when there is some sparing of the ability of rotation feedback to modulate PAN with changes in the orientation of the head with respect to gravity, one can envision some patients being able to decrease their PAN and improve vision by, for example lying down or lying on one side, such as when trying to read a book. When PAN is unaffected by gravity, however, it implies that the ability of the brain to counteract the destabilizing effects of virtual head rotation signals on an estimation of gravity has been neutralized.

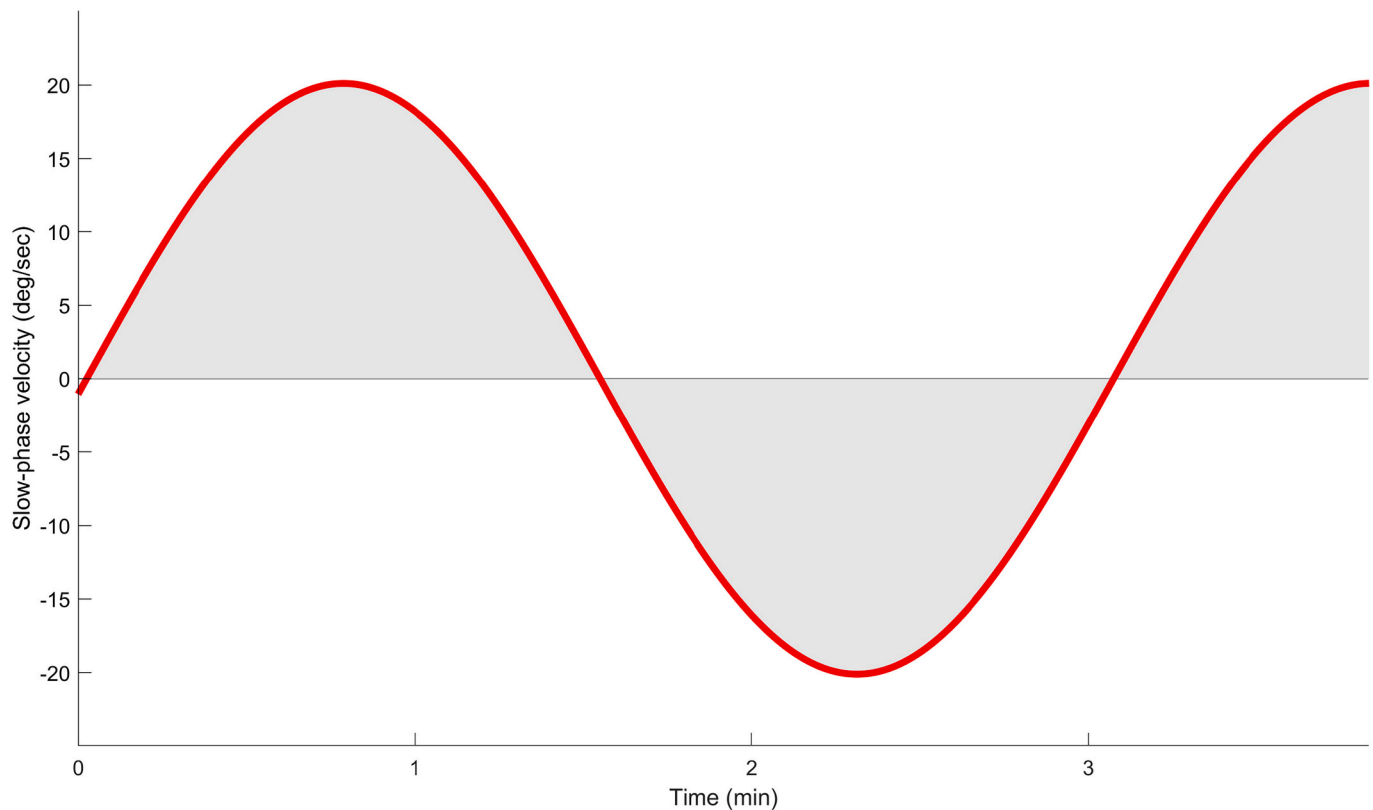


Fig. 5. Effect of left-ear down positioning on the pattern of PAN in the patient described by Furman et al. [2]. Horizontal slow-phase velocity (deg/s) is plotted against time (minutes). At time zero, the patient was brought promptly from sitting upright, to the left ear down position by tilting the head laterally 90 degrees about the roll axis. The solid red line is a simulation of the yaw axis angular velocity. Slow-phase velocity of the upright simulation is shown by the envelope of the gray shaded area. Parameter values for simulation include: PAN peak amplitude 20 deg./s, rotation feedback gain 0, and a bias of 0 deg./s. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.5. Limitations

One drawback of our model is that it ignores a potential contribution from impaired gaze-holding systems. When the eyes are held in an eccentric position in the orbit, an impaired gaze-holding system cannot maintain a constant position output to offset the orbital elastic forces that act like a spring to return the eye to its natural (null) position. This leads to centripetal drift that in turn, triggers a velocity command from the saccadic system to bring the eyes back to the desired eccentric position. A recurrent sequence of centripetal drift and a centrifugal saccade is the clinical signature of an impaired gaze-holding system, what is commonly called gaze-evoked nystagmus.

In patients with PAN who also have gaze-evoked nystagmus, slow-phase velocity is a linear combination of centripetal drift velocity (gaze-evoked nystagmus) and velocity signals arising from the unstable velocity-storage circuit [33]. The model used here, however, is insensitive to the position of the eyes in the orbit and therefore attributes changes in slow-phase velocity of the eyes only to modulation of PAN. Nevertheless, the general implications on PAN of altering head orientation with respect to gravity still apply.

In our model all the elements are symmetric and “lumped” that is, the model does not attempt to mimic the segregated laterality of the brain where there may be two different “rotation feedback” systems, one on each side of the brain. However, it is unlikely that lesions are perfectly symmetric and as such to fit the data properly we had to add a bias to the overall velocity to make the responses asymmetric. We do not know the source of this bias, though it is not unreasonable to assume such pathological biases exist and they can often explain complicated patterns of positional nystagmus [14].

5. Conclusions

What neuronal circuits could account for the rotation feedback gain effect on PAN in these patients? Our results could be consistent with known influences of the nodulus and ventral uvula of the cerebellum on vestibular function. Ablation of these structures in monkeys not only causes PAN, but also impairs tilt suppression of post-rotatory nystagmus, and eliminates the steady-state response during off-vertical axis rotation (OVAR). Together with the results of the ablations studies, we hypothesized that variations in the gain of rotation feedback account for the difference in the effects of gravity among PAN patients. In our patient, PAN was suppressed in our patient in ear-down positions, in a similar pattern to that of a previously reported patient. This effect was simulated by reducing the gain of the projection of the rotation feedback loop to the velocity-storage integrator to approximately 5% of its normal value. With normal rotation feedback PAN is expected to dissipate quickly as soon as the head is rotated away from upright position. Moreover, by disconnecting the rotation feedback completely (gain = zero) the model simulated PAN that was reported to be unaffected by gravity. We conclude

- (1) The slowly oscillating “virtual” head rotation signals of PAN are mathematically integrated in supine and ear-down positions leading to an erroneous estimate of the orientation of the head to gravity since the head is stationary.
- (2) In PAN, the rotation feedback loop, normally used to resolve the tilt-translation ambiguity, may malfunction.
- (3) A very low gain of rotation feedback can lead to suppression of PAN in head down positions, but the effect takes a few minutes.

- (4) Our model also predicts that a decrease in the amplitude of the GIA vector would have the same effect of gravity on PAN as a low gain of rotation feedback.
- (5) With absent “rotation feedback” PAN is unaffected by head down positions.
- (6) In patients with PAN, understanding how brain resolves the tilt-translation ambiguity and compensates for inherent imperfections of the ability to faithfully transduce the motion of the head, can enhance both the clinician’s ability to localize neurological disorders and our knowledge of how the normal brain functions.
- (7) The effects of changing the position of the head with respect to gravity on other forms of nystagmus due to a vestibular imbalance, allows one to interrogate the function of a normal or pathologically altered rotation feedback circuit.

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